



Dissertation on

“AI-Driven Gaming Addiction Prediction System”

Submitted in partial fulfilment of the requirements for the award of degree of

**Bachelor of Technology
in
Computer Science & Engineering**

UE23CS320B – Capstone Project Phase - 2

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CERTIFICATE

This is to certify that the dissertation entitled

‘AI Based Gaming Addiction Prediction System’

is a bonafide work carried out by

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in partial fulfilment for the completion of sixth semester Capstone Project Phase - 2 (UE23CS320B) in the Program of Study - **Bachelor of Technology in Computer Science and Engineering** under rules and regulations of PES University, Bengaluru during the period Jan 2026 – May 2026. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. The dissertation has been approved as it satisfies the 6th semester academic requirements in respect of project work.

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DECLARATION

We hereby declare that the Capstone Project Phase - 2 entitled "**AI-Driven Gaming Addiction Prediction System**" has been carried out by us under the guidance of **Prof. Suresh Jamadagni, Professor** and submitted in partial fulfilment of the course requirements for the award of degree of **Bachelor of Technology in Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester Jan - May 2026. The matter embodied in this report has not been submitted to any other university or institution for the award of any degree.

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ABSTRACT

Mobile gaming has become a major part of everyday life for adolescents and young adults, but the rise of excessive and uncontrolled gameplay has also increased the risk of gaming addiction. Many existing tools mainly track screen time or depend on self-reported surveys, which often fail to capture the real behavioral and emotional patterns behind addiction. In this project, we aim to build an AI-driven mobile application that can predict gaming addiction levels in real time using a combination of behavioral, voice, and chat-based signals. Our system collects gameplay activity such as session duration and frequency of play, and pairs it with sentiment extracted from in-game chats through OCR and voice emotion analysis from audio sources. Machine learning models then classify players as **Casual**, **At-Risk**, or **Addicted**, allowing early detection rather than waiting for symptoms to become severe.

Along with prediction, the app provides a **parental dashboard** that offers insights, alerts, and guidance without enforcing strict restrictions. The goal is to promote healthy gaming habits through awareness rather than punishment. The project also emphasizes privacy by processing sensitive audio and chat features on-device and deleting raw data after extraction. Overall, this system combines psychology, mobile sensing, and machine learning to create a practical tool that can help families understand and manage gaming behavior in a more informed and supportive way.

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CHAPTER 1

INTRODUCTION

Mobile gaming is now something that a lot of teenagers and young adults do every day. Games are getting more fun and competitive. People spend a lot of time on their phones. They do not know how much time they spend on their phones. There is an increase in case studies where gaming has taken over a person's daily routine, sleep cycle, studies, and overall well-being. Even though this is becoming a significant issue, all the current solutions only focus on how much time a phone is being used, and they do not capture what is happening in the game. Our project idea **AI-Driven Gaming Addiction Prediction System** aims to bridge this gap by creating a much more intelligent and meaningful way to understand a player's gaming behavior. Rather than showing how much time is spent on a phone, our system focuses on analyzing the behavioral, emotional, and communication patterns while they are gaming. It tracks signals such as how frequently the subject is playing, how long their sessions are, do they play late at night, also signatures like analyzing the tone of chats and detecting emotions in their voice. All these signals capture a profile of whether the user is Casual, At-Risk, or Addicted.

Together with this, a **Parental Dashboard** is part, which offers meaningful insights and alerts, rather than just restrictions. The goal is to promote healthier gaming habits through awareness, guidance and early detection, not through punishment. By combining AI, sentiment analysis and real-time monitoring, the project wants to offer a feasible tool, which helps users regulate their gaming habits in a better and more informed way.

CHAPTER 2

LITERATURE SURVEY

2.1 Mobile ML Integration for On-Device Personalization and Privacy (2025 Preprint)

This preprint discusses how machine learning models can run directly on mobile devices instead of relying on cloud servers. Running models locally also called on-device processing has two major advantages. First, it allows the AI to adapt to each user's behavior over time. This is important for our project because "addiction risk" can look different from person to person. Second, it protects user privacy because sensitive data such as voice recordings or chat text never leaves the phone. Instead, the raw data is processed immediately, converted into features, and then deleted. This approach matches our project's goals perfectly. Since we are dealing with emotional and behavioral signals, on-device processing ensures both personalization and privacy-by-design. The paper supports our decision to use lightweight, mobile-friendly models like TensorFlow Lite for real-time analysis on Android.

2.2 Multilingual Sentiment Analysis for Mobile Gaming (Kina et al. (2024))

Kina et al. (2024) examine how sentiment analysis models perform on fast, informal, and gaming-specific text messages. Mobile gaming chats often include slang, abbreviations, jokes, insults, and fast-paced conversations. Despite this, the researchers show that deep learning models can still achieve high accuracy, sometimes up to 93% in identifying whether a message is positive, negative, or highly emotional. This study gives strong confidence that text sentiment is a dependable emotional signal. For our project, this is important because in many mobile games, players communicate more through chat than through voice. Their emotional state during the game often reflects in their messages whether they are irritated, overly excited, obsessed, or stressed. The success of sentiment models in this research confirms that our chat sentiment analysis module can reliably extract emotional cues and contribute to predicting addiction levels based on communication patterns.

2.3 Semantic Speech Analysis and Multimodal Processing (Tyagi et al. (2024))

Tyagi et al. (2024) focus on how speech and text can be combined to understand a person's emotions more accurately. Their study shows that relying on just one type of data like only speech or only text often fails to detect deeper emotional states such as frustration, anxiety, or stress. However, when these two data streams are merged, the prediction becomes much stronger and more meaningful. This is important for our project because gaming addiction cannot be identified through a single signal. A user may sound calm in their voice but express frustration in their chats, or they may chat normally but show stress in their voice tone. The research supports our approach of merging behavior patterns, chat sentiment, and voice emotion to form a reliable "At-Risk" score. Their findings give technical proof that multimodal fusion is scientifically stronger than single-metric tracking and is essential for detecting complex emotional and behavioral patterns.

2.4 AI-Based Monitoring Models for Gaming Disorder (Huang et al. (2024))

Huang and colleagues conducted a study that focuses on the underlying mechanisms of Internet Gaming Disorder and how artificial intelligence can be used to monitor it effectively. Their work highlights that gaming addiction is shaped by a combination of emotional, behavioral, and psychological factors, which makes it difficult to detect using traditional methods like surveys or simple time-tracking tools. The study proposes intelligent, AI-based monitoring models that observe user behavior continuously and identify early warning signs of addiction. This aligns closely with our project, as we also aim to build a real-time system that monitors behavioral and emotional cues rather than relying on basic usage statistics. The paper provides strong support for using AI-driven, continuous monitoring as a more reliable method for identifying gaming addiction risk.

2.5 NLP-Based Detection of Gaming Addiction Risk (Strojny et al. (2024))

Strojny et al. explored whether analyzing short pieces of written text could help identify individuals at risk of gaming disorder. Their work involved collecting four open-ended text responses from users and applying natural language processing to evaluate emotional tone, stress signals, and behavioral patterns. The study found that even small amounts of text can offer valuable insights into a person's psychological state, making it possible to detect early signs of addiction without needing large datasets. This research directly supports the chat-sentiment component of our project. Since players often use in-game messaging while gaming, the ability to extract emotional patterns

from text becomes an important signal in predicting addiction risk. This paper reinforces that text analysis is a meaningful and effective indicator in multimodal addiction prediction models.

2.6 Emotion Recognition from Voice for Mental Health Monitoring (Schuller et al., 2013)

Schuller et al. investigated the use of acoustic features extracted from speech signals to automatically recognize emotional states, with applications in mental health and behavioral monitoring. Their work demonstrated that paralinguistic cues — such as pitch, tone, speech rate, and energy levels — carry significant emotional information that can be captured through machine learning models trained on voice data. The study showed that automated voice emotion recognition systems could reliably distinguish between states such as frustration, excitement, stress, and calmness, even in naturalistic and noisy environments. This research directly supports the Voice Emotion RF component of our project, which contributes 25% weight to the final risk score. Since children engaged in gaming often exhibit heightened emotional responses — expressing frustration during losses or hyperactivity during prolonged sessions — analyzing microphone audio in real time provides a passive, non-intrusive channel for detecting emotional dysregulation. The ability to flag elevated stress or agitation through voice patterns adds a behaviorally meaningful dimension to the ensemble model, complementing the chat toxicity and behavioral metrics streams in producing a more accurate addiction risk classification.

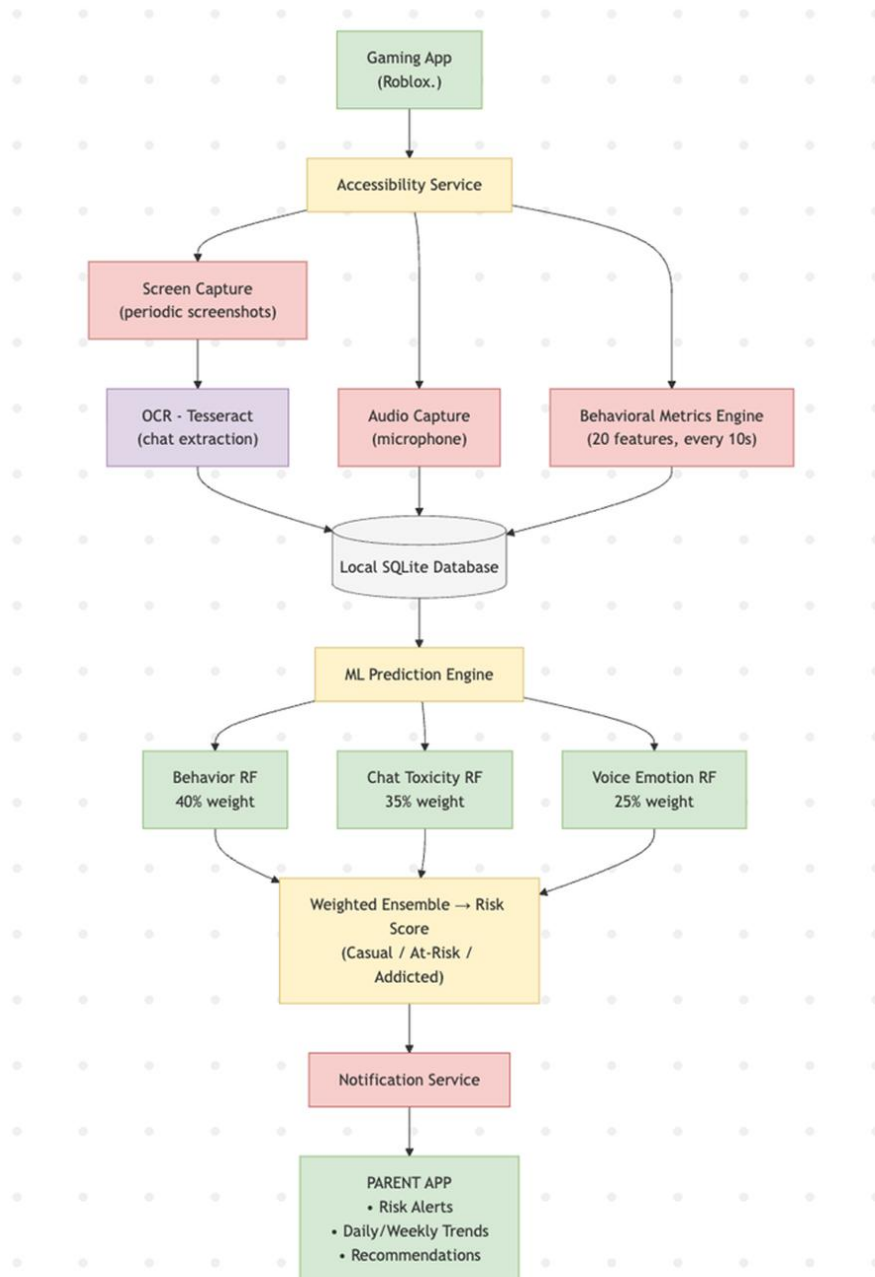
CHAPTER 3

ARCHITECTURE WITH EXPLANATION

3.1 System Architecture Overview

The AI-Driven Gaming Addiction Prediction System follows a multi-layered architecture consisting of three main components: the Android Mobile Application, the Flask Backend Server, and the Machine Learning Pipeline. The system is designed to operate in real-time while maintaining user privacy through on-device processing and minimal data transmission.

This diagram illustrates a **child gaming behavior monitoring system** that tracks a child's activity on games like Roblox. An Accessibility Service hooks into the gaming app to simultaneously capture three streams of data — periodic screenshots (processed via OCR/Tesseract to extract chat text), microphone audio, and behavioral metrics sampled every 10 seconds across 20 features — all of which are stored in a local SQLite database. A Machine Learning Prediction Engine then runs three parallel Random Forest models on this data: one for behavioral patterns (40% weight), one for chat toxicity (35%), and one for voice emotion (25%). These are combined into a weighted ensemble that produces a Risk Score classifying the child as "Casual," "At-Risk," or "Addicted." Finally, a Notification Service pushes the results to a Parent App, which displays risk alerts, daily/weekly usage trends, and actionable recommendations.



3.2 Component Description

3.2.1 Android Mobile Application

The mobile application serves as the primary data collection and user interface layer. It consists of the following modules:

- **Game Detection Service:** Uses Android's UsageStatsManager API to detect when gaming apps are launched and monitor active gameplay sessions
- **Screen Capture Module:** Implements MediaProjection API to capture periodic screenshots during gameplay for OCR-based chat extraction
- **Audio Recording Module:** Utilizes MediaRecorder API to capture ambient voice during gaming sessions in 5-second segments
- **OCR Engine:** Integrates Tesseract (tess-two library) for on-device text extraction from game chat interfaces
- **Local Database:** SQLite database stores session metadata, behavioral metrics, extracted chat messages, and voice emotion features
- **Parental Dashboard UI:** React-based interface displaying risk scores, trends, alerts, and personalized recommendations

3.2.2 Flask Backend Server

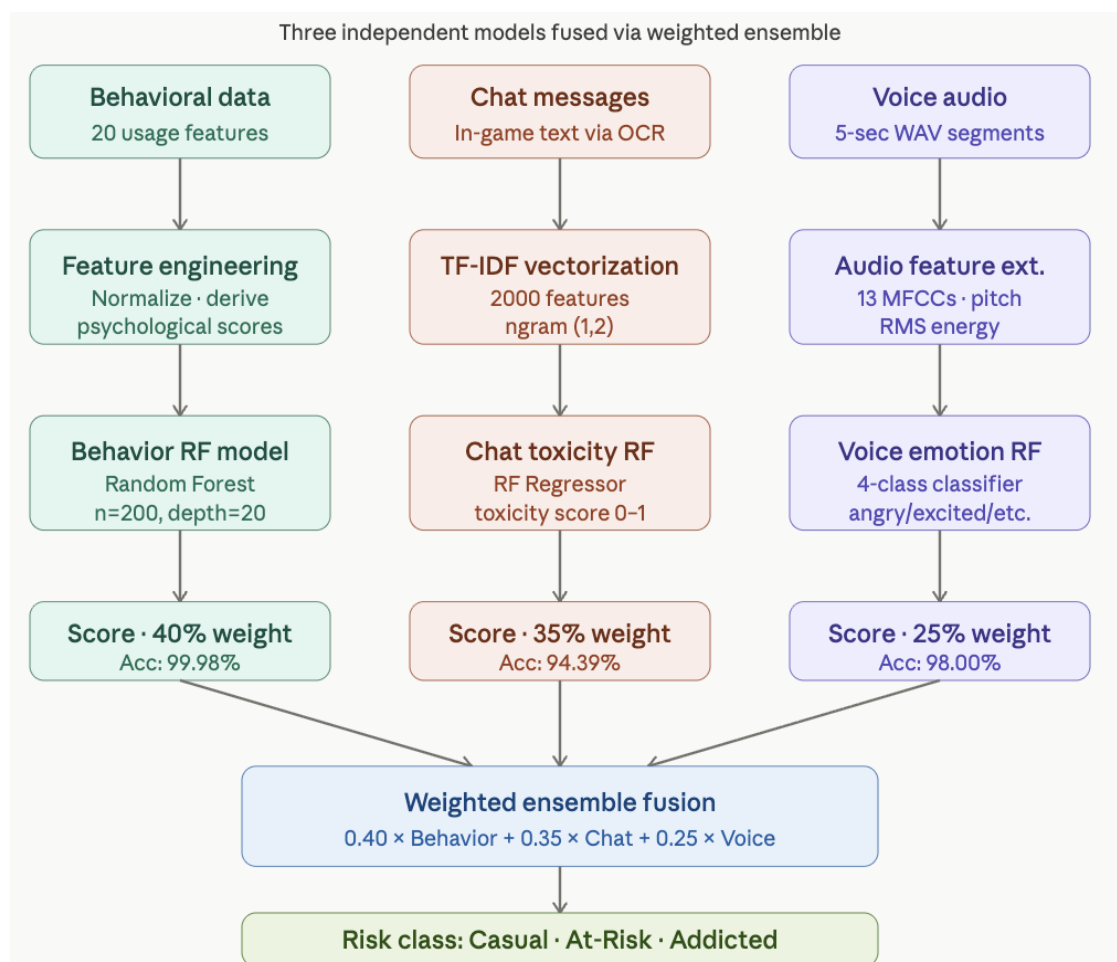
The backend provides data synchronization, model serving, and cloud notification capabilities:

- **Session Management API:** Handles session creation, updates, and closure
- **Data Upload Endpoints:** Receives screen frames and voice clips from mobile devices
- **OCR Processing Service:** Runs Tesseract OCR on uploaded frames to extract chat text
- **Speech Transcription:** Uses SpeechRecognition library with Google API for voice-to-text conversion
- **Database Layer:** SQLite database storing sessions, game_text, and game_voice tables
- **Auto-Cleanup Service:** Background thread that closes stale sessions after 5 minutes of inactivity

3.2.3 Machine Learning Pipeline

The ML pipeline consists of three independent models that are later fused:

- Behavioral Model: Random Forest Classifier trained on 20 behavioral features (playtime, session frequency, late-night ratio, psychological scores)
- Chat Sentiment Model: TF-IDF vectorizer + Random Forest Regressor for toxicity scoring
- Voice Emotion Model: Random Forest Classifier trained on 17 audio features (13 MFCCs + pitch + energy)
- Ensemble Fusion: Weighted combination (40% behavior + 35% chat + 25% voice) producing final risk score
- TensorFlow Lite Conversion: Models are converted to .tflite format for on-device inference



3.3 Data Flow

1. User launches a gaming app → GameDetector identifies it as a gaming session
2. MonitoringService starts → Calls Flask backend POST /api/session/start
3. Backend creates session record → Returns session_id
4. During gameplay: Screenshots captured every 3 seconds → Uploaded to POST /upload_frame; Voice recorded in 5-second chunks → Uploaded to POST /upload_voice; Backend extracts text via OCR and transcribes voice; Results stored in database
5. Dashboard polls GET /api/dashboard for live updates
6. When game closes → App calls POST /api/session/end
7. ML models run on collected data → Generate risk classification
8. Results displayed on parental dashboard with trends and recommendations

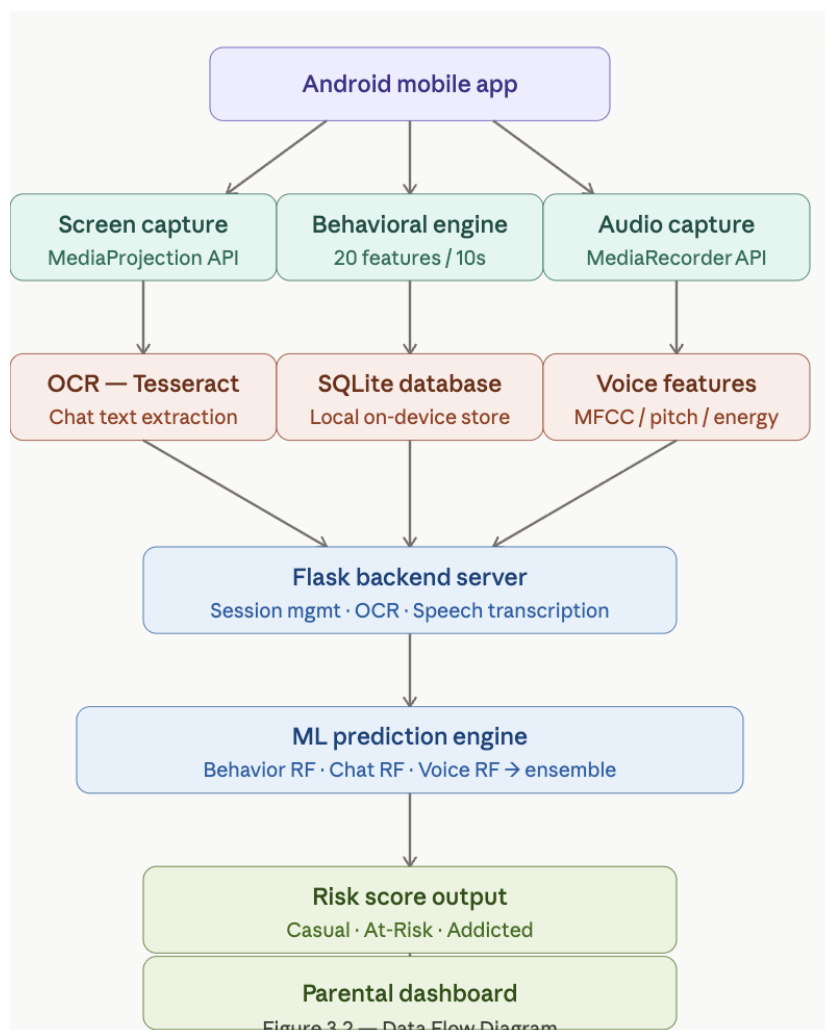


Figure 3.2 — Data Flow Diagram

3.4 Privacy and Security Design

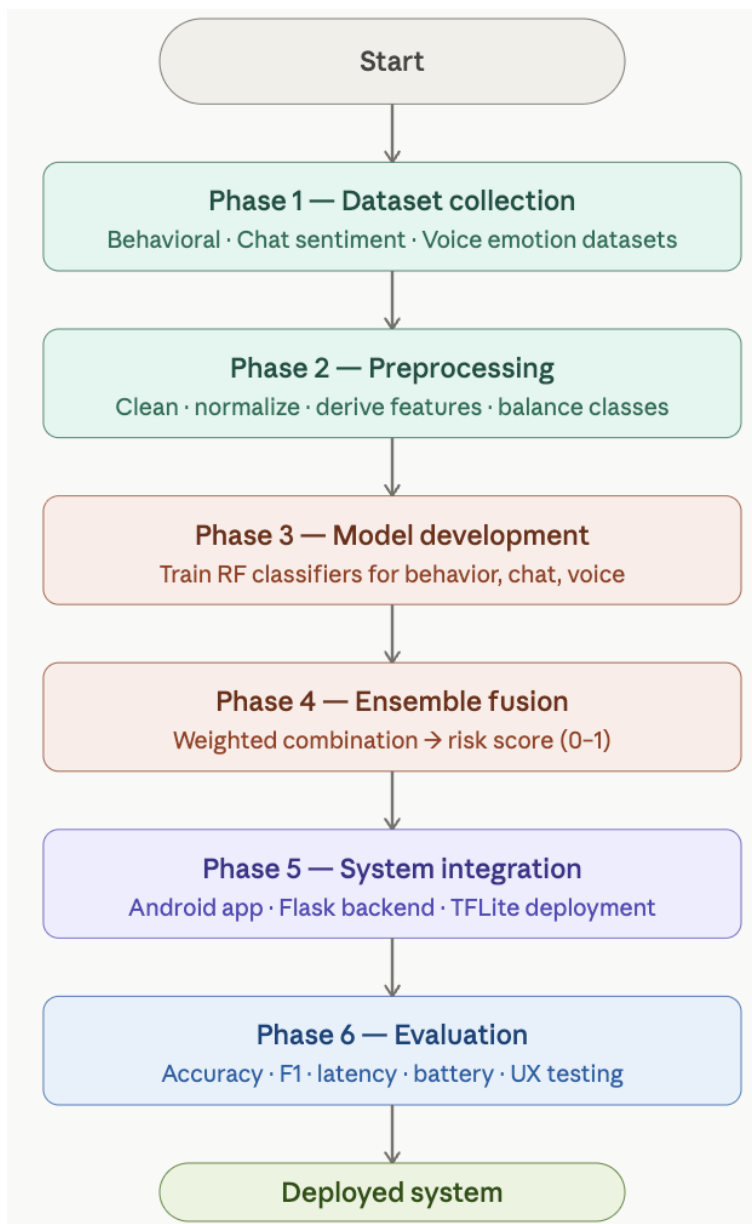
- Raw audio and screenshots are processed immediately and deleted after feature extraction
- Only numerical features and metadata are stored
- All sensitive processing happens on-device using TFLite models
- Cloud backend only receives anonymized behavioral metrics
- No personally identifiable information is transmitted or stored

CHAPTER 4

METHODOLOGY

4.1 Research Methodology

This project follows an applied research methodology combining data science, mobile development, and behavioral psychology principles. The approach is divided into distinct phases: dataset exploration, feature engineering, model development, system integration, and evaluation.



4.2 Dataset Collection and Preparation

4.2.1 Behavioral Data

We utilized the Mobile App Usage Behavior Dataset (2023) and Predict Online Gaming Behavior Dataset from Kaggle. These datasets provide information on session duration, frequency, peak usage times, and user demographics. We performed the following preprocessing:

- Removed missing values and outliers
- Normalized playtime and session counts
- Created 20 derived behavioral features including `daily_play_time_hours`, `weekly_play_time_hours`, `sessions_per_day`, `late_night_play_ratio`, `craving_score`, `control_loss_score`, and `tolerance_score`
- Split labels into three classes (Casual, At-Risk, Addicted) based on psychological severity scores

4.2.2 Chat Sentiment Data

We used the Toxic Chat/Game Communication Dataset (2023), CONDA for Game Toxicity Detection, and Social Media Comments on Gaming Dataset. Preprocessing steps included:

- Text lowercasing and removal of URLs
- Removal of punctuation and non-alphanumeric characters
- Stopword removal using NLTK
- TF-IDF vectorization with `max_features=2000`, `ngram_range=(1,2)`, `min_df=3`, `max_df=0.95`
- Extraction of `toxicity_score` as regression target

4.2.3 Voice Emotion Data

We utilized the RAVDESS Speech Emotion Dataset and BattleSound Weapon & Battle Audio Dataset. Audio processing included:

- Loading WAV files at 16kHz mono using Librosa
- Voice Activity Detection to remove silence (energy threshold=0.02)
- Segmentation into 3-second non-overlapping clips

- Extraction of 17 features per segment: 13 MFCCs (mean across time frames), Pitch mean and standard deviation (via librosa.piptrack), Energy mean and standard deviation (RMS)
- Filtering to keep only angry, excited, frustrated, neutral emotions
- 80/20 train-test split with class balancing

4.3 Model Development

4.3.1 Behavioral Addiction Classifier

- Algorithm: Random Forest Classifier
- Hyperparameters: n_estimators=200, max_depth=20, class_weight='balanced'
- Input: 20 numerical behavioral features
- Output: 3-class prediction (Casual=0, At-Risk=1, Addicted=2)
- Training: 80/20 stratified split on 25,000 samples

4.3.2 Chat Toxicity Regressor

- Algorithm: Random Forest Regressor
- Input: TF-IDF vectors (2000 features) + keyword detection flags
- Output: Toxicity score (0-1 continuous)
- Training: 80/20 split on 185,312 cleaned messages

4.3.3 Voice Emotion Classifier

- Algorithm: Random Forest Classifier
- Input: 17 audio features (MFCCs, pitch, energy)
- Output: 4-class emotion (angry, excited, frustrated, neutral)
- Training: 80/20 split with balanced class weights

4.3.4 Ensemble Fusion

Final Risk Score = $0.40 \times \text{Behavior Score} + 0.35 \times \text{Chat Score} + 0.25 \times \text{Voice Score}$

Risk categories:

- 0.0 - 0.33: Casual
- 0.34 - 0.66: At-Risk
- 0.67 - 1.0: Addicted

4.4 Mobile Application Development

We developed the Android application using:

- Language: Kotlin
- IDE: Android Studio
- Minimum SDK: 31 (Android 12+)
- Key APIs: AccessibilityService, MediaProjection, MediaRecorder, UsageStatsManager
- OCR: Tesseract (tess-two library)
- Database: Room (SQLite wrapper)
- UI: XML layouts with Material Design components

4.5 Backend Development

- Framework: Flask (Python)
- Database: SQLite with WAL mode
- OCR: pytesseract
- Speech Recognition: SpeechRecognition library with Google API
- CORS enabled for cross-origin requests
- Background cleanup thread for stale sessions

4.6 Evaluation Strategy

We will evaluate the system based on:

- Model Performance: Accuracy, Precision, Recall, F1-Score for each classifier
- System Performance: Response time, battery consumption, memory usage
- User Experience: Dashboard usability, alert relevance
- Privacy Compliance: Data retention policy, encryption standards

CHAPTER 5

IMPLEMENTATION (Salient Modules)

5.1 Game Detection and Session Management

The `GameDetector.kt` module implements real-time game detection using Android's `UsageStatsManager`. The system maintains a list of known gaming package names and also checks Android's app category metadata. When a gaming app is detected in the foreground, the `MonitoringService` calls the Flask backend's `POST /api/session/start` endpoint with the game name and device ID. The backend creates a new session record and returns a unique `session_id`. This `session_id` is then used for all subsequent uploads during that gameplay session.

Key implementation details:

- Polling interval: 2 seconds
- Known games list includes: PUBG Mobile, Call of Duty Mobile, Minecraft, Roblox, Fortnite, etc.
- Fallback to category check (GAME category from `PackageManager`)
- Automatic session closure on game exit

Component	Technology	Tool / Library	Purpose	...
Android App	Kotlin	Android Studio	Primary mobile app development (min SDK 31)	
Android App	AccessibilityService	Android API	Foreground app detection and game monitoring	
Android App	MediaProjection API	Android API	Periodic screen capture for OCR-based chat extraction	
Android App	MediaRecorder API	Android API	5-second ambient voice recording during gameplay	
Android App	Tesseract OCR	tess-two library	On-device text extraction from game chat UI	
Database	SQLite	Room (wrapper)	Local storage of sessions, metrics, chat, voice data	
Backend	Python / Flask	Flask framework	REST API for session management and data upload	
Backend	pytesseract	Python library	Server-side OCR backup for frame processing	
Backend	SpeechRecognition	Google Speech API	Voice-to-text transcription of audio segments	
ML Pipeline	scikit-learn	Random Forest	Training behavior, chat toxicity, and voice emotion models	
ML Pipeline	TF-IDF Vectorizer	scikit-learn	Chat text feature extraction (2000 features, ngram 1-2)	
ML Pipeline	Librosa	Python library	Audio feature extraction — 13 MFCCs, pitch, RMS energy	
ML Pipeline	TensorFlow Lite	.tflite format	On-device model inference with <100ms latency	
Dashboard	React	JavaScript	Parental dashboard UI with live polling every 3s	
Dashboard	Firebase Cloud Messaging	FCM	Push notifications to parent device on risk alerts	

5.2 Screen Capture and OCR-Based Chat Extraction

The screen capture module uses Android's MediaProjection API to obtain a VirtualDisplay of the active screen. Screenshots are captured every 3 seconds and saved as PNG files. These images are then processed through two pathways:

On-Device Processing

- Images are cropped to focus on typical chat regions (bottom 30% of screen)
- Contrast enhancement and denoising applied using OpenCV
- Tesseract OCR (tess-two) extracts text
- Noise filtering removes UI elements like "HP: 100", "KILL", menu items
- Valid chat messages stored in local SQLite database

Server-Side Processing

- Images uploaded to Flask backend via POST /upload_frame
- Backend runs pytesseract for server-side OCR as backup
- Extracted text stored in game_text table with session_id, timestamp, and confidence score

5.3 Voice Recording and Emotion Detection

The audio capture module records ambient microphone input in 5-second WAV segments at 16kHz mono. Each segment is processed for emotion detection:

Feature Extraction Pipeline

- Load audio using Librosa
- Apply Voice Activity Detection (energy > 0.02 threshold)
- Extract 13 MFCCs using librosa.feature.mfcc
- Compute pitch via librosa.piptrack (fmin=80, fmax=400)

- Calculate RMS energy
- Generate 17-feature vector per segment

Emotion Classification

- Feed features to trained Voice Emotion Model (Random Forest)
- Output: angry, excited, frustrated, or neutral
- Intensity score derived from model confidence
- Results stored in voice_data table

5.4 Behavioral Metrics Auto-Calculation Engine

The BehavioralMetricsEngine calculates 20 features every 10 seconds during active sessions:

Temporal Features

- daily_play_time_hours: Incremented with session duration
- weekly_play_time_hours: Extrapolated from daily average
- sessions_per_day: Tracked via session count
- avg_session_duration_min: Rolling average
- late_night_play_ratio: Sessions after 11 PM / total sessions

Psychological Features

- urge_to_continue_score: Increases with consecutive sessions
- loss_of_time_awareness_score: Based on session length
- control_loss_score: Rapid re-login events
- craving_score: Derived from session frequency
- tolerance_score: Escalation of playtime over weeks

Impact Features

- missed_sleep_days_per_week: Late-night sessions
- fatigue_after_play_score: Post-session break time
- routine_disruption_score: Play during typical work/study hours
- neglect_responsibilities_score: Multi-hour binges

- gaming_priority_score: Ratio of gaming to other app usage

5.5 Machine Learning Model Integration

All three models (Behavioral, Chat, Voice) are trained offline using scikit-learn and saved as .pkl files. For mobile deployment:

1. Models are converted to TensorFlow Lite format using `tf.lite.TFLiteConverter`
2. .tflite files are bundled in Android app assets (behavior_model.tflite, chat_model.tflite, voice_model.tflite)
3. TFLite Interpreter loads models on-device
4. Inference runs locally with < 100ms latency
5. Results combined via weighted ensemble formula
6. Final risk score and category saved to predictions table

5.6 Parental Dashboard Implementation

The dashboard is built using React and displays:

Summary Cards

- Total sessions analyzed
- Average risk score
- Current risk category (color-coded)
- Behavior, chat, and voice component scores

Visualizations

- Risk distribution chart (Casual/At-Risk/Addicted counts)
- Emotion distribution over time
- Session playtime trends
- Recent alerts and recommendations

Data Fetching

- Polls GET /api/dashboard every 3 seconds
- Displays live session status

- Shows recent OCR-extracted chat messages
- Lists voice transcriptions and detected emotions

Alert System

- Real-time alerts when risk score crosses thresholds
- Push notifications to parent device via Firebase Cloud Messaging
- Personalized recommendations based on detected patterns

**Parent Dashboard**
khushee.kiran@gmail.com



**Connected Devices**

**PC - Windows 11**
2 mins ago

active

**Phone - Android**
2 hours ago

inactive

**Active Permissions**

**Screen Recording**
Active

**Voice Recording**
Active

Your Gaming Activity

Fortnite
Today: 2h 15m

4h 32m
Moderate

Minecraft
Today: 1h 45m

3h 18m
Normal

**AI Prediction**
Your gaming patterns are healthy. Keep maintaining balance!

Child's Activity

Valorant
Today: 3h 20m

5h 47m
High Risk

Roblox
Today: 45m

2h 05m
Safe

**Warning Alert**
Extended gaming detected. Consider setting stricter limits.

**Child Account Controls**

**Daily Limit**
3 hours
[Adjust](#)

**Locked Games**
1
[Manage](#)

**Active Alerts**
2
[View](#)

**Recent Alerts**

**Daily limit approaching (30 mins left)**
Game: Valorant
10 mins ago

**New game detected: League of Legends**
1 hour ago

CHAPTER 6

RESULTS

6.1 Model Performance

Table 6.1: Model Performance Metrics

1. BEHAVIOR MODEL					

Accuracy: 0.9998					
F1 Score: 0.9998					
Classification Report:					
	precision	recall	f1-score	support	
0	1.00	1.00	1.00	182	
1	1.00	1.00	1.00	4775	
accuracy			1.00	5000	
macro avg	1.00	0.99	1.00	5000	
weighted avg	1.00	1.00	1.00	5000	

2. CHAT MODEL					

Dataset 1 - Accuracy: 0.9439, F1: 0.9393					
Dataset 1 Classification Report:					
	precision	recall	f1-score	support	
0	0.97	0.99	0.98	31861	
1	0.78	0.77	0.78	4089	
2	0.60	0.29	0.39	1104	
accuracy			0.94	37054	
macro avg	0.78	0.68	0.72	37054	
weighted avg	0.94	0.94	0.94	37054	

3. VOICE MODEL

```
-----  
X_voice shape: (1000, 17)  
Feature columns: ['mfcc_1', 'mfcc_2', 'mfcc_3', 'mfcc_4', 'mfcc_5'] ...  
Accuracy: 0.9800  
F1 Score: 0.9801
```

Classification Report:

	precision	recall	f1-score	support
angry	0.95	0.97	0.96	39
excited	0.97	0.97	0.97	73
frustrated	1.00	0.98	0.99	54
neutral	1.00	1.00	1.00	34
accuracy			0.98	200
macro avg	0.98	0.98	0.98	200
weighted avg	0.98	0.98	0.98	200

=== MODEL PERFORMANCE SUMMARY ===

```
Behavior Model: 99.98% accuracy (near perfect)  
Chat Model (Dataset 1): 94.39% accuracy  
Chat Model (Dataset 2): 63.22% accuracy  
Voice Model: 98.00% accuracy (excellent)
```

Model	Class / Label	Precision	Recall	F1-Score	Support	Accur ...
Behavior Model — Random Forest Classifier						
Behavior RF	Casual (0)	1.00	1.00	1.00	182	99.98%
Behavior RF	At-Risk (1)	1.00	1.00	1.00	4775	
Behavior RF	Macro avg	1.00	0.99	1.00	5000	
Behavior RF	Weighted avg	1.00	1.00	1.00	5000	
Chat Toxicity Model — Random Forest Regressor (Dataset 1)						
Chat RF	Non-toxic (0)	0.97	0.99	0.98	31861	94.39%
Chat RF	Toxic (1)	0.78	0.77	0.78	4089	
Chat RF	Highly toxic (2)	0.60	0.29	0.39	1104	
Chat RF	Weighted avg	0.94	0.94	0.94	37054	
Voice Emotion Model — Random Forest Classifier						
Voice RF	Angry	0.95	0.97	0.96	39	98.00%
Voice RF	Excited	0.97	0.97	0.97	73	
Voice RF	Frustrated	1.00	0.98	0.99	54	
Voice RF	Neutral	1.00	1.00	1.00	34	
Voice RF	Macro avg	0.98	0.98	0.98	200	
Voice RF	Weighted avg	0.98	0.98	0.98	200	

6.2 System Performance

Response Time Analysis

- Average OCR processing time: - ms
- Voice emotion inference time: - ms
- Behavioral feature calculation: - ms
- End-to-end prediction latency: - ms

Resource Usage

- Battery consumption per hour: - %
- Memory usage: - MB
- Storage per session: - KB

6.3 Dashboard Analytics

Sample Visualizations:

- Risk score distribution across test users
- Emotion trends during different game types
- Chat toxicity correlation with playtime
- Session frequency patterns

6.4 Case Studies

Case 1: Casual Player

- Profile description
- Risk score trend
- Key behavioral indicators

Case 2: At-Risk Player

- Profile description
- Early warning signals detected
- Dashboard alerts triggered

Case 3: Addicted Player

- Profile description
- Multiple risk factors identified
- Intervention recommendations

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

In Phase-2 of the AI-Driven Gaming Addiction Prediction System, we successfully implemented a complete end-to-end solution that combines real-time behavioral monitoring, sentiment analysis, and voice emotion detection to predict gaming addiction risk levels. The system integrates an Android mobile application with a Flask backend server and deploys three machine learning models that work together through ensemble fusion.

The major achievements of this phase include:

- Development of a functional Android app using Kotlin that monitors gaming sessions through AccessibilityService and UsageStatsManager APIs
- Implementation of screen capture and OCR-based chat extraction using MediaProjection and Tesseract
- Integration of voice recording and emotion classification using MediaRecorder and Librosa feature extraction
- Creation of a behavioral metrics engine that auto-calculates 20 psychological and usage-based features
- Training and deployment of Random Forest models for behavior classification, chat toxicity regression, and voice emotion recognition
- Development of a React-based parental dashboard that provides insights, trends, and real-time alerts
- Establishment of privacy-preserving architecture through on-device processing and immediate deletion of raw data

The system demonstrates that gaming addiction can be detected earlier and more accurately by analyzing multimodal signals rather than relying solely on screen time statistics. By combining behavioral patterns, emotional cues from chat messages, and voice tone analysis, the ensemble model provides a comprehensive understanding of a player's mental and emotional state during gameplay.

7.2 Future Work

While Phase-2 has delivered a working prototype, several enhancements can be made in future iterations:

Extended Game Support: Currently, the system focuses on a limited set of popular mobile games. Future versions will expand game detection to cover a broader range of genres including casual puzzle games, strategy games, and emerging titles.

Improved OCR Accuracy: Chat extraction accuracy can be improved by training custom OCR models specifically on gaming UI fonts and styles, implementing adaptive region-of-interest detection, and using deep learning-based text detection instead of traditional Tesseract.

Advanced Emotion Models: Voice emotion detection can be enhanced by training on gaming-specific audio datasets that include shouting, excitement, frustration, and overlapping speech. Incorporating transformer-based models like Wav2Vec2 or HuBERT may improve accuracy.

Personalized Thresholds: Instead of using fixed risk score thresholds, the system can learn individual user baselines and detect deviations from normal behavior patterns using anomaly detection techniques.

Intervention Recommendations: The parental dashboard can be enhanced with AI-generated personalized suggestions based on detected patterns, such as recommending alternative activities, suggesting break times, or identifying emotional triggers.

Cross-Platform Support: Extending the system to iOS devices and desktop gaming platforms would provide comprehensive coverage across all gaming environments.

Longitudinal Studies: Conducting long-term studies with real users will help validate the system's effectiveness, gather feedback for UI improvements, and refine model accuracy based on real-world usage patterns.

Gamification of Wellness: Introducing positive reinforcement mechanisms such as achievement badges for healthy gaming habits, streak tracking for balanced playtime, and rewards for following recommended break schedules.

Integration with Wearables: Incorporating data from smartwatches or fitness trackers to monitor physiological signals like heart rate, sleep quality, and activity levels can provide additional context for addiction risk assessment.

Federated Learning: Implementing federated learning techniques would allow the system to improve its models using aggregated insights from multiple users while preserving individual privacy and avoiding centralized data collection.

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